

Internship in Physical Organic Chemistry

Week 1 — Part 3: Reading papers using the reading template

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Goals:

- Work through each selected paper using the reading template
- Distinguish clearly between observation and interpretation
- Extract the information you will need to write your Background chapter
- Understand what each paper contributes and how papers relate to each other

In Part 2 you selected 3 to 5 papers. In this part you will read them carefully and fill in a copy of the reading template for each one. The reading template is not a form to fill in quickly — it is a thinking tool. Working through it forces you to be precise about what a paper actually shows and what it claims.

By the end of this part, each paper in your selection should have a completed template. These templates are your working notes: they will be the basis for everything you write in Part 4.

1 How to use the reading template

The reading template (`reading_template.pdf`, provided separately) has four sections. You do not complete all sections for every paper. Instead:

- **Section A** is completed for every paper, without exception.
- **Section B** is completed if the paper reports experimental measurements or spectra (tick *Experimental* or *Mixed* in Step 1 of the template).
- **Section C** is completed if the paper is a review or overview.
- **Section D** is completed if the paper reports calculations, energy profiles, or theoretical reasoning.

Many papers are mixed: a paper that reports both spectra and a mechanistic argument supported by calculations requires both Section B and Section D. Tick all boxes that apply in Step 1.

Before you start reading. Print or open a blank copy of the template for each paper. Write the paper number at the top (e.g. Paper 1, Paper 2, ...) so that your templates stay organised. Do not fill in all templates at once — work through one paper at a time, from start to finish.

2 Section A: what every paper must tell you

Section A asks you to record the basic bibliographic information and to identify the main contribution of the paper in your own words.

The most important part of Section A is the question: *what is the main question or topic of this paper?* Answer this in two or three sentences, in your own words. Do not copy the abstract.

If you cannot answer this question without looking at the paper, read the abstract again and try once more.

Task. For each paper, complete Section A before reading further. In particular:

- Write the bibliographic information *from Zotero*, not from memory. Check that author names, year, journal name, and title are correct.
- Write the main topic in your own words. One or two sentences is enough.
- List three key things the paper tells you. These do not need to be in complete sentences at this stage.
- Decide how you will use this paper. Tick all boxes that apply in the last part of Section A.

The question about how you will use the paper forces you to think about the paper's role *in your report*, not just its content. A paper can be interesting without being useful for your Background chapter. Knowing the difference now will save time later.

3 Section B: experimental papers

Section B is the most important section for most of the papers you will read in this project. It asks you to separate what was *observed* from what was *interpreted*.

This distinction is fundamental to scientific reasoning and is often harder to maintain than it appears. The exercises in this section train you to make it consistently.

3.1 Observation versus interpretation

An **observation** is something directly measured or detected: a peak at a given mass-to-charge ratio, a change in intensity, a neutral loss of a specific mass. It is a statement about data. It can be verified by looking at the figure.

An **interpretation** is a proposed explanation of why that observation occurs: a bond broke, a rearrangement took place, a particular pathway was followed. It is a statement about mechanism. It cannot be verified by looking at the figure alone.

Illustrative example. Consider the following two statements:

1. "A fragment ion was detected at m/z 153."
2. "Loss of the B-ring occurred via retro-Diels-Alder cycloreversion."

Statement 1 is an observation. It describes what appears in the spectrum and can be verified directly from the figure.

Statement 2 is an interpretation. It proposes a mechanism that is consistent with the observation, but is not itself directly visible in the spectrum.

Both statements are valid and necessary. The error is to present statement 2 as if it were statement 1.

When reading a paper, authors often mix observation and interpretation within the same sentence. Part of your task is to separate them even when the paper does not.

Task. For each experimental paper, complete Section B as follows:

1. Identify the three most important *observations* in the paper. These must be things you can see directly in a figure. Write them as plain statements of what was detected or measured.
2. For each observation, write the interpretation the authors give. If the authors do not give one explicitly, note this.
3. Answer the question at the bottom of Section B: can you clearly distinguish observation from interpretation in this paper? Tick the box that applies honestly.

3.2 Reading figures

In an experimental paper about fragmentation, the most important figures are the mass spectra and the fragmentation schemes. Spend time on these before reading the text.

When looking at a mass spectrum, identify the precursor ion (usually the largest mass shown or the most intense peak at the left), the base peak (the most intense fragment), and any labelled peaks. Then read the caption. Captions often explain what the figure shows without requiring you to read the full results section.

Once you have a sense of what the spectrum looks like, read the section of the paper that discusses it. This is usually in the Results or Results and Discussion section. Find the sentences where the authors describe the fragments you identified in the figure.

Task. For each experimental paper, identify the figure that contains the most relevant fragmentation data. Record in Section B which figure it is and what it shows. If the paper contains a fragmentation scheme (a diagram showing proposed bond cleavages), note this separately: it is an interpretation, not a measurement.

4 Section C: review papers

If any of your selected papers is a review, Section C asks you to extract vocabulary and identify references worth following up.

The primary purpose of a review paper in your literature collection is to give you language. Reviews define terms, introduce notation, and describe the field's conventions. When you read a review, the most valuable output is a list of terms and their definitions.

Task. For each review paper, complete Section C as follows:

1. List five terms or concepts that were new to you, or that were used differently than you expected. For each term, write a one-line definition in your own words.
2. Identify two to four references from the review that seem directly relevant to your topic. Record author(s), year, and a one-sentence note on why each is worth reading.
3. Extract three to five search terms you could use to find additional papers. These should be chemistry terms, not instrument names.

Do not attempt to read all the papers cited in a review. A review may cite fifty or a hundred papers. Your task is to identify a small number that are directly connected to your project and that you would not have found through your initial search.

5 Section D: computational papers

If any of your selected papers reports calculations, Section D asks you to identify the method, the quantities computed, and how these connect to experiment.

At this stage you are not expected to understand the calculations in detail. The goal is to read the paper as a consumer of its results: what did the authors calculate, what did they find, and how do they use these results to explain or predict experimental behaviour?

Task. For each computational paper, complete Section D as follows:

1. Write the name of the computational method used (e.g. DFT, semi-empirical, molecular mechanics). If several methods are used, list them all.
2. Identify the main quantities reported. Common examples: energy barriers, optimised geometries, orbital energies, reaction profiles. For each quantity, write what it represents in physical terms.
3. Find the place in the paper where computational results are compared to experimental data. Write one sentence explaining how the authors use the calculation to support their experimental interpretation.

Note. You will return to computational papers in Week 4, when you carry out your own calculations. At that point, you will be able to evaluate these papers in much more detail. For now, the goal is only to understand what kind of information calculations provide and how they relate to the spectra you are studying.

6 Comparing your papers

Once you have completed a template for each paper, step back and look at your set of templates together. This comparison is as important as reading each paper individually.

Task. With your completed templates in front of you, answer the following questions in your lab notebook or Overleaf document. Write two to four sentences for each.

1. **What do your papers agree on?** Identify one or two observations or interpretations that appear in more than one paper. Write exactly what they agree on.
2. **Where do your papers differ?** Identify one case where two papers describe similar data but reach different conclusions, or where one paper discusses something the others do not. Write what the difference is and why it might matter.
3. **What is missing?** Is there a question raised by one paper that none of the others answers? Is there a compound, condition, or comparison that would be useful but is not in your collection?

This comparison exercise produces the first draft of an argument. You are not yet writing prose, but you are identifying what the literature says as a whole, not just what each individual paper says. This is what a Background chapter does: it synthesises, it does not summarise.

Illustrative example. After reading three papers on fragmentation of a class of compounds, you might find that:

1. All three report the same two major fragment ions under low-energy CID. This suggests these fragments are diagnostic and worth describing in your Background.
2. Two papers assign one fragment to a specific bond cleavage, while the third assigns it differently. This is a point of genuine uncertainty in the literature — it should be noted in your Background, not hidden.
3. None of your papers reports what happens at higher collision energies. This connects directly to what you will measure in Week 2.

7 What to do if you are stuck

Reading scientific papers is a skill that develops with practice. At this stage, it is normal to find certain papers difficult or to be unsure about specific assignments.

If you cannot identify the main observation in a figure, try reading the caption first, then the first and last sentence of the paragraph that discusses the figure. These two sentences usually contain the most important information.

If you are unsure whether a statement is an observation or an interpretation, ask yourself: could I verify this directly from the figure, without any knowledge of chemistry? If yes, it is an observation. If no, it is an interpretation.

If a paper seems entirely unclear or unrelated to your topic after working through Section A, it may not belong in your selection. Set it aside and note why. You do not need to force a paper into your collection if it does not fit.

End of Part 3 checklist. Before moving to Part 4, verify that:

- You have a completed reading template for each of your 3–5 selected papers.
- Every template has Section A filled in fully.
- Every experimental paper has Section B filled in, with observations and interpretations listed separately.
- You have written answers to the three comparison questions in Section 5.
- You can say in one sentence what each paper contributes that the others do not.

If any of these is missing, complete it before continuing. Part 4 builds directly on the work done here.